HTML Notes: Chapter 5 – Images

Adding images, choosing and storing them, size, placement, captions

Basics

- Reasons to add an image: logo, photograph, illustration, diagram or chart.
- Taking time to get images right makes a site look more attractive and professional.
- This chapter: include an image using HTML, pick the image format, show an image at the right size, and optimize images so pages load faster.
- You can also add images with CSS using the `background-image` property.

Choosing images for your site

- Great images make the difference between an average-looking site and a really engaging one.
- Images set the tone for a site faster than reading a description.
- No photos of your own? Companies sell **stock images** you pay to use.
- All images are subject to **copyright**. You can get into trouble for taking photos from another website.
- For a page with several images (product photos, team members), a simple, consistent background makes them look better as a group.
- **Images should:** be relevant; convey information; convey the right mood; be instantly recognisable; fit the colour palette.

STOCK PHOTO WEBSITES (from the book)

```
www.istockphoto.com
www.gettyimages.com
www.veer.com
www.sxc.hu
www.fotolia.com
```

Storing images on your site

- Building a site from scratch? Good practice: make one folder for all the images the site uses.
- As a site grows, a separate folder helps you understand how the site is organized.
- On a big site, add subfolders inside the images folder, e.g. **interface** (logos, buttons), **products** (product photos), **news** (news images).
- With a CMS or blogging platform there are usually tools in the admin site to upload images, and the program probably already has a separate folder for them.

EXAMPLE SITE FILES

```
From A to Zucchini/
  about.html
  contact.html
  images/
    chocolate-islands.jpg
    lemon-posset.jpg
    logo.gif
    roasted-brussel-sprouts.jpg
    zucchini-cake.jpg
  index.html
```

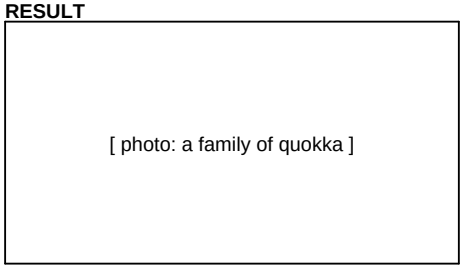
Adding images `<img>`

- `<img>` adds an image. It is an **empty element** (no closing tag).
- `src` = where the browser finds the image file. Usually a relative URL to an image on your own site (here, in a child folder called images).
- `alt` = text description of the image, for when you cannot see it. Called **alt text**. It should accurately describe the image so screen readers (used by people with visual impairments) and search engines understand it.
- `title` = extra information about the image. Most browsers show it as a tooltip when you hover over the image.
- If the image is just decoration with no meaning (e.g. a graphic dividing line), still use `alt` but leave the quotes **empty**.

HTML

```
<img src="images/quokka.jpg"
  alt="A family of quokka"
  title="The quokka is an
  Australian marsupial that
  is similar in size to the
  domestic cat." />
```

RESULT



Hovering shows the title as a tooltip.

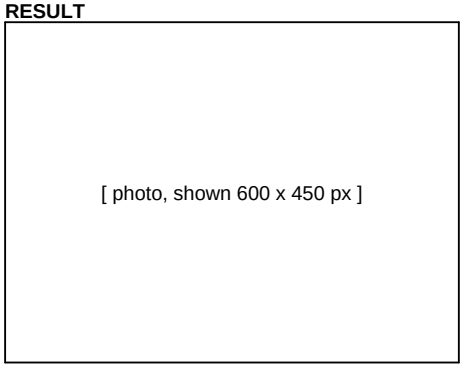
Height & width of images

- `height` = height of the image in pixels. `width` = width of the image in pixels.
- Images often take longer to load than the HTML around them. Giving the size lets the browser draw the rest of the text and leave the right amount of space for the image still loading.
- Image size is increasingly set with CSS instead of HTML.

HTML

```
<img src="images/quokka.jpg"
  alt="A family of quokka"
  width="600" height="450" />
```

RESULT



Where to place images in your code

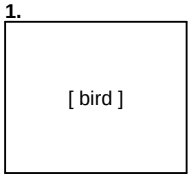
- Where the image sits in the code changes how it is displayed. Browsers show elements in two ways:
- **Block elements** always start on a new line, e.g. `<h1>` and `<p>`.
- **Inline elements** sit inside a block element and do not start on a new line, e.g. `<b>`, `<em>` and `<img>`.
- **1. Before a paragraph:** the paragraph is a block element, so it starts on a new line after the image.
- **2. Inside the start of a paragraph:** the image is inline, so the first row of text lines up with the bottom of the image.
- **3. In the middle of a paragraph:** the image sits between the words of the paragraph and text flows around it.

HTML

```
<img src="images/bird.gif"
  alt="Bird" width="100"
  height="100" />
<p>There are around 10,000
  living species of birds...</p>
<hr />
<p><img src="images/bird.gif"
  alt="Bird" width="100"
  height="100" />There are
  around 10,000 living species
  of birds...</p>
<hr />
<p>There are around 10,000
  living species of birds.<img
  src="images/bird.gif"
  alt="Bird" width="100"
  height="100" />Many species
  undertake long distance...</p>
```

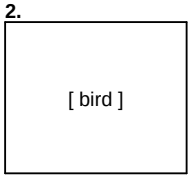
RESULT

1.



There are around 10,000 living species of birds that inhabit different ecosystems from the Arctic to the Antarctic.

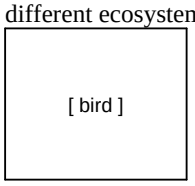
2.



There are around 10,000 living species of birds that inhabit different ecosystems from the Arctic to the Antarctic.

3.

There are around 10,000 living species of birds that inhabit different ecosystems.



Many species undertake long distance annual migrations.

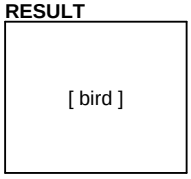
Old code: aligning images horizontally `align`

- The `align` attribute was commonly used to make other content flow around an image.
- **Removed from HTML5.** New sites should use CSS (covered later). It's here because you'll see it in older code and some visual editors still add it.
- `left`: image on the left, text flows around its right-hand side.
- `right`: image on the right, text flows around its left-hand side.
- Looks neater than one line of text beside the image.
- Text right up against the image is harder to read. Use the CSS `padding` and `margin` properties to add a gap.

HTML

```
<p><img src="images/bird.gif"
  alt="Bird" width="100"
  height="100" align="left" />
  There are around 10,000
  living species of birds...</p>
<hr />
<p><img src="images/bird.gif"
  alt="Bird" width="100"
  height="100" align="right" />
  There are around 10,000
  living species of birds...</p>
```

RESULT



There are around 10,000 living species of birds that inhabit different ecosystems from the Arctic to the Antarctic.

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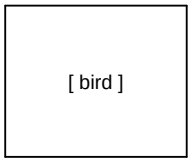


Old code: aligning images vertically `align`

- Also no longer used in HTML5, but you may see it in older sites and in code from some visual editors. CSS does the same job.
- `top`: first line of text lines up with the top of the image; other lines go under the image.
- `middle`: first line lines up with the vertical middle of the image; other lines go under.
- `bottom`: first line lines up with the bottom of the image; other lines go under.
- To wrap **all** the text around the image (not just one line), use the CSS `float` property.
- In older code, `align` with `left` or `right` gave the same wrap effect, but it's no longer recommended.

HTML

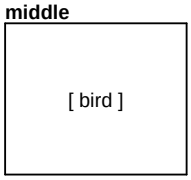
```
<p><img src="images/bird.gif"
  alt="Bird" width="100"
  height="100" align="top" />
  There are around 10,000...</p>
<hr />
<p><img ... align="middle" />
  There are around 10,000...</p>
<hr />
<p><img ... align="bottom" />
  There are around 10,000...</p>
```

RESULT

top

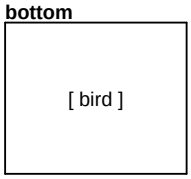
There are around 10,000 living species of birds that inhabit different ecosystems from the Arctic to the Antarctic.

middle



There are around 10,000 living species of birds that inhabit different ecosystems from the Arctic to the Antarctic.

bottom



There are around 10,000 living species of birds that inhabit different ecosystems from the Arctic to the Antarctic.

HTML5: figure and caption `<figure>` `<figcaption>`

- Images often come with captions. HTML5 added `<figure>` to hold an image and its caption so the two are associated.
- You can have more than one image inside `<figure>` as long as they share the same caption.
- Browsers sometimes indent the contents of `<figure>`.
- `<figcaption>` = the caption for the image. Before these elements there was no way to link an `<img>` to its caption.
- Older browsers that don't understand HTML5 elements ignore them and just show the content.

HTML

```
<figure>
  <img src="images/otters.jpg"
    alt="Photograph of two
    sea otters floating
    in water" />
  <br />
  <figcaption>Sea otters hold
    hands when they sleep so
    they don't drift away from
    each other.</figcaption>
</figure>
```

RESULT

Sea otters hold hands when they sleep so they don't drift away from each other.

Quick cheat sheet

Tag / attribute	Use
`<img />`	Add an image (empty element)
`src`	Where the image file is (relative URL)
`alt`	Text description (leave empty for decorative images)
`title`	Tooltip text on hover
`width / height`	Image size in pixels
`block element`	Starts on a new line (h1, p)
`inline element`	Stays in the line (b, em, img)
`align (old)`	left/right/top/middle/bottom, removed in HTML5, use CSS
`<figure>`	Holds an image and its caption
`<figcaption>`	The caption